Cluster 9744

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Subjects

Research fields: Mathematics, Physics, Computer science
The most common research fields of articles in this cluster. Read more.

Research subfields: Applied mathematics, Control theory, Mathematical analysis
Common research subfields of articles in this cluster. Read more.

Key concepts: Aerodynamic Shape Optimization, Adjoint Method, Shape Optimization Problems, Discrete Adjoint Solver, continuous adjoint
These key concepts are identified algorithmically using article titles and abstracts. Read more.

1.68% of all articles are AI-related
Articles are algorithmically labeled as AI-related or not. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. Read more.

0.42% of English-language articles are related to robotics
Articles are labeled as robotics-related or not using an ETO model. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. Read more.

0.00% of English-language articles are related to computer vision
Articles are labeled as computer vision-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. Read more.

0.00% of English-language articles are related to natural language processing
Articles are labeled as NLP-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. Read more.

0.00% of English-language articles are related to AI safety
Articles are labeled as related to AI safety or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. Read more.

0.00% of English-language articles are related to cybersecurity

Vital signs

241 recent articles in the cluster
241 articles in this cluster were published in the five year period ending in 2023.

Average article is **16.54** years old
The average age of articles in this cluster. However, most Map of Science metrics are calculated using articles from the past five years only.

Growth rating: **4.33th** percentile
Over the three year period ending in 2023, this cluster grew faster than roughly 4.33% of other clusters.

Citation rating: **45.52th** percentile
A cluster's citation rating is equal to the average citation percentile for all the articles in the cluster. Citation percentiles are calculated relative to articles of the same publication year.In this case, the average article in this cluster is cited more often than roughly 45.52% of all other articles published in the same year.

94.61% of articles are in English

Not expected to grow extremely fast
As determined by ETO's growth prediction model. Read more.

Countries and languages

Top author countries

Each row lists the number of articles in this cluster from the last five years with at least one author from an organization in the specified country. Data available for 214 of 241 articles. 182 of these articles had at least one author from an organization in these ten countries.

Country	Articles (five year period ending in 2023)	Yearly citations per article (average)
United States	41	1.09
China	39	1.15
Germany	35	1.44
Greece	25	0.76
United Kingdom	22	0.97
India	12	0.14
Canada	11	0.42
Spain	10	0.57
Italy	9	1.16
France	9	0.67

Top funding countries

Data available for 67 of 241 articles. 60 of these articles disclosed funding from one or more organizations in these ten countries.

Country	Articles (last five years)	Yearly citations (average)
China	23	1.85
Germany	10	2.47
United States	8	1.40
Belgium	5	0.86
United Kingdom	5	0.46
Greece	4	1.88
Canada	3	1.70
Spain	2	3.45
Netherlands	2	3.70
Switzerland	1	3.50

Articles and sources

Core articles

Core articles are especially highly connected to other papers within the cluster.

- [Aerodynamic shape optimization based on discrete adjoint and RBF]
  2022: SSRN Electronic Journal. 14 citations.
- [Multistage Turbomachinery Design Using the Discrete Adjoint Method Within the Open-Source Software SU2]
  2020: Journal of Propulsion and Power. 27 citations.
- [Analysis of finite-volume discrete adjoint fields for two-dimensional compressible Euler flows]
  2020: arXiv. 9 citations.
- [Multi-Row Turbomachinery Aerodynamic Design Optimization by an Efficient and Accurate Discrete Adjoint Solver]
  2023: Aerospace. 8 citations.
- [Continuous adjoint-based shape optimization of a turbomachinery stage using a 3D volumetric parameterization]
  2023: International Journal for Numerical Methods in Fluids. 9 citations.
- [A review on aerodynamic optimization of turbomachinery using adjoint method, 2024]
  2024: Part C: Journal of Mechanical Engineering Science. 3 citations.
- [Analytic adjoint solutions for the 2-D incompressible Euler equations using the Green's function approach]
  2022: arXiv. 4 citations.

Review articles

Review articles have between 100 and 1000 out-citations. These articles describe and systematize other contributors' research.

- [On the Contribution of Wall Distance Fields to the Adjoint of a RANS Model]
  2022: International journal of computational fluid dynamics. 1 citations.
- [A review on aerodynamic optimization of turbomachinery using adjoint method, 2024]
  2024: Part C: Journal of Mechanical Engineering Science. 3 citations.

Highly cited articles

Highly cited articles are the articles in the cluster with the most citations overall.

- [Proper orthogonal decomposition assisted inverse design optimisation method for the compressor cascade airfoil]
  2020: Aerospace Science and Technology. 28 citations.
- [Multistage Turbomachinery Design Using the Discrete Adjoint Method Within the Open-Source Software SU2]
  2020: Journal of Propulsion and Power. 27 citations.
- [Numerical investigation of minimum drag profiles in laminar flow using deep learning surrogates]
  2020: arXiv. 31 citations.
- [Aerodynamic Shape Optimization]
  2022: Computational Aerodynamics. 39 citations.
- [Turbomachinery, 2020]
  2020: Sixth Edition. 65 citations.
- A novel p-harmonic descent approach applied to fluid dynamic shape optimization
  2021: arXiv: Optimization and Control. 29 citations.
- [Shape optimization to improve the transonic fluid-structure interaction stability by an aerodynamic unsteady adjoint method]
  2020: Aerospace Science and Technology. 22 citations.

Top sources

The most common sources of articles from the last five years in this cluster, such as specific journals, conference proceedings, or preprint servers. Data available for 234 out of 241 articles. 76 of these articles came from these sources.

Source	Articles (last five years)
AIAA Aviation 2019 Forum	10
arXiv	10
Computers & Fluids	9
AIAA SCITECH 2022 Forum	9
AIAA Journal	8
Aerospace Science and Technology	7
International Journal for Numerical Methods in Fluids	7
AIAA AVIATION 2021 FORUM	6
Aerospace	5
arXiv (Cornell University)	5

Authors, organizations, and funders

Top authors

The authors with the most articles from the last five years in this cluster. Data available for 240 of 241 articles. 47 of these articles were written by these authors.

Author	Articles (last five years)	Yearly citations (average)
Kyriakos C. Giannakoglou	23	0.93
E.M. Papoutsis-Kiachagias	14	0.53
Xenofon Trompoukis	11	1.16
E.M. Papoutsis-Kiachagias	10	0.54
Varvara Asouti	9	1.06
Dingxi Wang	8	3.56
Κωνσταντίνος Τσιάκας	8	1.09
Thomas Rung	8	9.73
Jorge Ponsín	7	0.57
Niklas Kuehl	7	3.01

Top author organizations

The author organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 221 of 241 articles. 81 had at least one author from one of these ten organizations.

Organization	Articles (last five years)	Yearly citations (average)
National Technical University of Athens - Greece	25	0.76
Northwestern Polytechnical University - China	12	1.64
Rheinland-Pfälzische Technische Universität Kaiserslautern-Landau - Germany	11	1.68
Universität Hamburg - Germany	8	2.64
Hamburg University of Technology - Germany	8	2.64
Instituto Nacional de Técnica Aeroespacial - Spain	7	0.52
National Institute of Aerospace - United States	7	0.83
Politecnico di Milano - Italy	6	1.39
Xi'an Jiaotong University - China	6	1.07
Von Karman Institute for Fluid Dynamics - Belgium	5	0.95

Organization types

This table covers the 208 articles in this cluster from the five year period ending in 2023 with data about their author organization types (out of 241 articles total). Read more

Commercial	Education	Nonprofit	Government
5.59%	42.20%	6.95%	2.03%

Top industry organizations

The industry organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 25 out of 241 articles. 17 of these articles had at least one author from an organization listed below.

Organization	Articles (last five years)	Yearly citations (average)
Ansys (United States)	4	0.79
BMW Group (Germany)	3	2.33
Scientific Simulations (United States)	2	0.4
Volkswagen Group (United States)	2	1.1
Robert Bosch (United States)	2	0.8
Software (Germany)	2	0.17
Dassault Aviation (France)	1	0.25
Embraer (Brazil)	1	1.6
Mitsubishi Electric (United States)	1	1.67
ESI Group (France)	1	0.25

Top funders

Data available for 95 out of 241 recent articles. 44 of these articles disclosed funding from one or more of these organizations.

Funding organization	Articles (last five years)	Yearly citations (average)
National Natural Science Foundation of China	17	1.71
Ministry of Science and Technology - China	10	2.30
National Major Science and Technology Projects of China	6	3.31
Federal Ministry for Economic Affairs and Climate Action - Germany	5	2.44
H2020 Marie Skłodowska-Curie Actions	5	0.80
Deutsche Forschungsgemeinschaft - Germany	5	3.62
National Science and Technology Major Project - China	5	1.02
Hellenic Foundation for Research and Innovation - Greece	4	1.88
INTA	4	0.60
National Science Foundation - United States	4	2.06

Funder types

This table covers the 63 articles in this cluster from the last five years with data about their funding organization types (out of 241 articles total).

Commercial	Education	Nonprofit	Government
3.90%	7.79%	15.58%	72.73%

Patents and industry

46 patents cite articles in this cluster (**66.44th** percentile among all clusters)

1.37% of articles in the cluster are cited by patents

5.09% of articles have at least one author from industry

Top patent titles

Patent title	Same-cluster citations
A STREAMING COMPILER FOR AUTOMATIC ADJOINT DIFFERENTIATION	2
AUTONOMOUS SLAT-COVE-FILLER DEVICE FOR REDUCTION OF AEROACOUSTIC NOISE ASSOCIATED WITH AIRCRAFT SYSTEMS	2
OPTIMIZATION OF AN AUTOMATICALLY MESHABLE SHAPES	2
METHODS FOR DIRECTED SELF-ASSEMBLY PROCESS/PROXIMITY CORRECTION	1
OPTIMIZATION OF THE PHYSICAL PROPERTIES OF A SUBMERGED OR FLOWN-THROUGH GEOMETRIC BODY	1
CURVED WING TIP	1
SPLIT BLENDED WINGLET	1
ADJOINT-BASED CONDITIONING OF PROCESS-BASED GEOLOGIC MODELS	1
TRANSFER LINE EXCHANGER	1

Top patent assignees

Patent assignee	Same-cluster citations
ROHDE & SCHWARZ (GERMANY)	3
UNITED STATES OF AMERICA AS REPRESENTED BY THE ADMINISTRATOR OF THE NATIONAL AERONAUTICS AND SPACE ADMINISTRATION	2
AIRBUS (FRANCE)	2
EXXONMOBIL (UNITED STATES)	2
MATLOGICA LIMITED	2
AVIATION PARTNERS INC	2
GOLOUBENTSEV DMITRI	2
SIDUS INNOVATIONS OY	2
SUN TAO	1

Connections

Top citing clusters

Cluster	Number of significant citations	CSET extracted phrases
42963	102	Aerodynamic Shape Optimization, multidisciplinary design optimization, shape optimization design, aerodynamic design, airfoil shape design
28389	24	aerodynamic shape optimization, design space, airfoil aerodynamic design, robust optimization design, optimization design method
44645	16	Topology optimization, heat transfer, heat sink, microchannel heat sink, optimization method
31656	11	shape optimization problems, shape derivative, shape gradient, optimization problems constrained, PDE-constrained shape optimization
18778	11	Automatic Differentiation, Reverse-Mode Automatic, differentiable programming, adjoint, derivatives
23021	11	anisotropic mesh adaptation, Mesh adaptation methods, metric-based anisotropic mesh, discontinuous Galerkin methods, Adaptive Mesh
8977	10	harmonic balance method, blade flutter stability, blade vibration, aerodynamic, turbine blade cascade
34209	8	Machine learning, flow field, deep reinforcement learning, neural networks, convolutional neural network
2986	4	Discontinuous Galerkin Method, high-order Discontinuous Galerkin, Galerkin methods, flux reconstruction method, Galerkin finite element
9412	4	shape optimization, finite element method, optimization method, structural shape, proposed method

Top cited clusters

Cluster	Number of significant citations	CSET extracted phrases
42963	26	Aerodynamic Shape Optimization, multidisciplinary design optimization, shape optimization design, aerodynamic design, airfoil shape design
18778	16	Automatic Differentiation, Reverse-Mode Automatic, differentiable programming, adjoint, derivatives
23021	10	anisotropic mesh adaptation, Mesh adaptation methods, metric-based anisotropic mesh, discontinuous Galerkin methods, Adaptive Mesh
16737	10	mesh deformation method, Radial basis functions, function mesh deformation, basis function mesh, computational fluid dynamics
28389	8	aerodynamic shape optimization, design space, airfoil aerodynamic design, robust optimization design, optimization design method
5936	7	Mach number flows, incompressible fluid flow, computational fluid dynamics, flow simulations, turbulent flows
8977	7	harmonic balance method, blade flutter stability, blade vibration, aerodynamic, turbine blade cascade
9412	7	shape optimization, finite element method, optimization method, structural shape, proposed method
2465	6	Ahmed body, drag reduction, aerodynamic drag, wake flow, square-back Ahmed body
6423	5	rotor aerodynamic noise, broadband noise, interaction noise, Propeller Tonal Noise, acoustic

Top GitHub repositories

Repository	Number of within-cluster citations	Number of stars
KratosMultiphysics/Kratos	2	1027
pcarruscag/FADO	1	7